Complete proofs: kernel of a block, corank formula and duality between the blocks k and $n-k$

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for the family $y^n = x^d + \sigma x^r + 1$ over $\mathbb{F}_p$

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Abstract

The three points that the previous note left as observations (completeness of the formula corank $= m - \operatorname{rank} A$, bounds on m, duality law (K)) are proved here, by hand, for σ such that f is squarefree. The points not proved are listed at the end (cycles, $\gcd(d,n) > 1$, σ with f not squarefree).

0. Conventions

p prime, $p \nmid n$, $\gcd(d,n) = 1$, $1 \le r < d$, $\sigma \in \mathbb{F}_p^*$, $f = x^d + \sigma x^r + 1$ *squarefree over* $\overline{\mathbb{F}}_p$. We work over $K = \overline{\mathbb{F}}_p$ (the rank of a matrix with coefficients in $\mathbb{F}_p$ does not change under extension). $\mathcal{V}_m$ = polynomials of degree $< m$.

For $1 \le k < n$: $k_{\text{out}} \equiv kp^{-1} \pmod n$, $g_k = \lfloor kd/n \rfloor$, $N = (k_{\text{out}}p - k)/n \in [0, p-1]$, $j = p - 1 - N$. The block is $M^{(k)}[b,i] = [x^{pb-i}]f^N$, $1 \le b \le g_{k_{\text{out}}}$, $1 \le i \le g_k$. Write $g = g_k$ and $L(G) = fG' - jf'G$.

The kernel of $M^{(k)}$ is seen in $\mathcal{V}_g$ via $q \mapsto Q = \sum q_i x^{i-1}$: then $(Mq)_b = [x^{pb-1}](f^N Q)$.

1 The kernel consists of exact differentials

Lemma 1. *For $Q \in \mathcal{V}_g$ and $b > g_{k_{\text{out}}}$, $[x^{pb-1}](f^N Q) = 0$.*

Proof. If this coefficient is non-zero, $pb - 1 \le Nd + g - 1$, so $pb \le Nd + g = \frac{k_{\text{out}}pd - kd}{n} + g \le \frac{k_{\text{out}}pd}{n}$ because $g \le kd/n$. Hence $b \le k_{\text{out}}d/n$, i.e. $b \le g_{k_{\text{out}}}$. □

Thus $Mq = 0 \iff$ no term x^{pb-1} ($b \ge 1$) in $f^N Q \iff C(f^N Q\,dx) = 0$, where C is the Cartier operator ($C(x^{pb-1}dx) = x^{b-1}dx$, $C(x^m dx) = 0$ if $p \nmid m+1$, $C(h^p\omega) = hC(\omega)$, and $\ker C = d\,K(x)$).

Theorem 2 (kernel)**.** $\ker M^{(k)} \cong \{G \in \mathcal{V}_{dj} :\ \deg L(G) \le g-1\}$, *via* $G \mapsto Q = L(G)$.

Proof. Since $N = p - 1 - j$, $f^N = f^p f^{-(j+1)}$, so $C(f^N Q\,dx) = f \cdot C\big(Q\,dx/f^{j+1}\big)$; the kernel is therefore the set of $Q \in \mathcal{V}_g$ such that $\omega = Q\,dx/f^{j+1}$ is exact.

($\Leftarrow$) If $G \in \mathcal{V}_{dj}$, $d(G/f^j) = \frac{fG' - jf'G}{f^{j+1}}dx$; so $Q = L(G)$ gives an exact form.

($\Rightarrow$) Let $\omega = dh$, $h \in K(x)$. Decompose $h = P(x) + \sum_\beta \sum_{e \ge 1} c_{\beta,e}(x-\beta)^{-e}$. Then $dh = P'dx - \sum e\,c_{\beta,e}(x-\beta)^{-e-1}dx$: the terms with $p \mid e$ and the monomials of P of degree divisible by p disappear. Now $\omega = Q/f^{j+1}$ is a proper fraction ($\deg Q < d$), whose partial-fraction

decomposition has poles only at the roots α_i of f (simple), of order $\le j+1$. By uniqueness: $P'=0$, and $e\,c_{\beta,e}=0$ as soon as β is not a root or $e\ge j+1$. Put $\tilde h=\sum_i\sum_{1\le e\le j}c_{\alpha_i,e}(x-\alpha_i)^{-e}$. Since $j\le p-1$, no $e\le j$ is divisible by p, so $d\tilde h=dh=\omega$. The roots being simple, $\tilde h=G/f^j$ with G a polynomial, and $\tilde h\to 0$ at infinity gives $\deg G<dj$. Finally $d\tilde h=L(G)\,dx/f^{j+1}$ gives $Q=L(G)$. □

Lemma 3 (injectivity of L)**.** *L is injective on $\mathcal{V}_{dj}$.*

Proof. $L(G)=0\iff (G/f^j)'=0\iff G/f^j\in K(x^p)$. A p-th power has poles only of order a multiple of p; but G/f^j has poles of order $\le j<p$ at the roots of f: it has none, so it is a polynomial P^p. But $\deg G<dj$ gives $\deg(G/f^j)<0$, hence $P=0$, $G=0$. □

Corollary 4. $\dim\ker M^{(k)}=\dim W$, *where* $W=\{G\in\mathcal{V}_{dj}:\deg L(G)<g\}$.

2 The corank formula (completeness)

The Laurent polynomials $\Phi_M=x^{-pM}f^j$ ($M\ge 1$) satisfy $L(\Phi_M)=0$: indeed $\Phi'_M=jx^{-pM}f^{j-1}f'$ (because $p\mid pM$), so $f\Phi'_M=jf'\Phi_M$. Write $\Phi_M=h_M+P_M$ (part with exponents ≥ 0 / part with exponents ≤ -1). Let $m=\lfloor dj/p\rfloor$; $h_M\ne 0\iff M\le m$, and $\deg h_M=dj-pM$.

Lemma 5. *For $M\le m$: $h_M\in\mathcal{V}_{dj}$, $L(h_M)=-L(P_M)$ has degree $\le d-2$.*

Proof. $L(h_M)=-L(P_M)$ because $L(\Phi_M)=0$. Now $L(x^t)=(t-jd)x^{t+d-1}+\sigma(t-jr)x^{t+r-1}+t\,x^{t-1}$ and P_M has only exponents $t\le -1$: the exponents of $L(P_M)$ are $\le d-2$. □

Proposition 6. $S:=\{G\in\mathcal{V}_{dj}:\deg L(G)\le d-2\}$ *has basis* $(h_M)_{1\le M\le m}$; $\dim S=m$.

Proof. The h_M ($M\le m$) lie in S (Lemma 5) and have distinct degrees: they are independent. Conversely let $G\in S$ be non-zero of degree s ($s\le dj-1$). The coefficient of x^{s+d-1} in $L(G)$ is $(s-jd)G_s$ (the other terms have exponents $\le s+r-1<s+d-1$); since $s+d-1\ge d-1>d-2$, we have $(s-jd)G_s=0$, so $s\equiv dj\pmod p$, i.e. $s=dj-pM$ with $1\le M\le m$. A subspace whose non-zero elements have degrees in a set of m values has dimension $\le m$. Hence $\dim S=m$ and $S=\mathrm{span}(h_M)$. □

Theorem 7 (corank formula)**.** *Let $A=(A_{M,e})$, $1\le M\le m$, $g\le e\le d-2$, $A_{M,e}=[x^e]L(P_M)$. Then* $\dim\ker M^{(k)}=m-\mathrm{rank}\,A$.

Proof. Since $g\le d-1$, $W\subset S$. For $G=\sum\lambda_M h_M$, $[x^e]L(G)=-\sum_M\lambda_M A_{M,e}$ for $g\le e\le d-2$, and $\deg L(G)\le d-2$; so $G\in W\iff \lambda A=0$. The h_M being independent, $\dim W=m-\mathrm{rank}A$. Conclude by Corollary 4. □

Corollary 8. *(i)* $\dim\ker M^{(k)}\ge m-\nu$, *where ν is the maximum matching number of the support graph of A (rank $\le$ term rank, König–Egerváry). (ii) In particular an identically zero row of A produces a kernel vector.*

3 Duality between k and $n-k$

Lemma 9 (arithmetic)**.** *For the block $n-k$: (a) $N(n-k)=p-1-N=j$; (b) $g_{n-k}=d-1-g$; (c) $(n-k)_{\mathrm{out}}=n-k_{\mathrm{out}}$ and $g'_{\mathrm{out}}:=g_{n-k_{\mathrm{out}}}=d-1-g_{k_{\mathrm{out}}}$; (d) $R:=\lfloor(dj+d-1-g)/p\rfloor=g'_{\mathrm{out}}$; (e) $m\le R$.*

Proof. (a) $\frac{(n-k_{\mathrm{out}})p-(n-k)}{n}=p-1-\frac{k_{\mathrm{out}}p-k}{n}$. (b),(c) $kd/n\notin\mathbb{Z}$, so $\lfloor d-kd/n\rfloor=d-1-g_k$. (d) Put $u=dk_{\mathrm{out}}\bmod n$, $v=dk\bmod n$ ($u,v\ne 0$); $up\equiv v\pmod n$. One computes $dj+d-1-g=p(d-g_{k_{\mathrm{out}}})-p\frac{u}{n}-(1-\frac{v}{n})$, whence $R=d-1-g_{k_{\mathrm{out}}}+\lfloor\theta\rfloor$ with $\theta=1-\frac{u}{n}-\frac{1-v/n}{p}$. Now $\theta<1$, and $p\,n\,\theta=pn-pu-n+v=n(p-1-c)$ with $c=(pu-v)/n\in\mathbb{Z}$, $c<pu/n<p$: so $\theta\ge 0$ and $\lfloor\theta\rfloor=0$. (e) obvious. □

Write $B = M^{(n-k)}$: by (a)–(c), $B[b,t] = [x^{pb-t}]f^j$ for $1 \le b \le R$, $1 \le t \le d-1-g$. Write $\kappa = \dim\ker M^{(k)}$ and $\gamma = \dim\operatorname{coker} B$ (dimension of the left kernel).

Theorem 10 (law (K))**.** $\kappa = \gamma$.

Proof. Let $X = K^R$; for $\mu \in X$ put $\Phi_\mu = \sum_{b=1}^{R} \mu_b \Phi_b$ (Laurent polynomial, $L(\Phi_\mu) = 0$), $\Phi_\mu = G(\mu) + P(\mu)$ (parts ≥ 0 and ≤ -1). Let $g' = d-1-g$.

Two linear maps. (1) $C : X \to K^{g'}$, $C_t(\mu) = [x^{-t}]\Phi_\mu = \sum_b \mu_b B[b,t]$, so $C(\mu) = \mu B$ and $\dim\ker C = \gamma$. (2) $E : X \to K^{g'}$, $E_e(\mu) = [x^e]L(G(\mu))$ $(g \le e \le d-2)$.

Link. $L(G) = -L(P)$ gives $E_e(\mu) = -[x^e]L(P(\mu))$. By the formula for $L(x^{-t})$, this coefficient depends only on the C_t with $t \le d-1-e \le g'$: $E = T \circ C$ (possibly after reindexing), in particular $\ker C \subset \ker E$. Explicitly, with $t = d-1-e$ and $t' = r-1-e$,

$$E_e(\mu) = (t + jd)\, C_t(\mu) + \sigma\,(t' + jr)\, C_{t'}(\mu) \qquad (C_{t'} := 0 \text{ if } t' < 1),$$

with positive signs (checked numerically on 784 random cases).

Computation of κ. By Proposition 6, $\mu \mapsto G(\mu)$ maps X onto S, with kernel $X_{>m} = \{\mu : \mu_b = 0\ \forall b \le m\}$ (dimension $R-m$). By Theorem 7 and its proof, W is the image of $\ker E$. Hence $\kappa = \dim\ker E - (R-m)$.

Claim: $\ker E = \ker C + X_{>m}$. Both spaces lie in $\ker E$ (for $b > m$, $h_b = 0$, $G(e_b) = 0$). Conversely let $\mu \in \ker E$, $t_1 = \min\{t \ge 1 : [x^{-t}]\Phi_\mu \ne 0\}$; if $t_1 > g'$, then $\mu \in \ker C$. Otherwise, $e_1 = d-1-t_1 \in [g, d-2]$ and the coefficient of x^{e_1} in $L(P(\mu))$ is obtained by $L(x^{-t}) = (-t-jd)x^{d-1-t} + \sigma(-t-jr)x^{r-1-t} - t\,x^{-t-1}$: it equals $-(t_1+jd)[x^{-t_1}]\Phi_\mu$, the other terms having strictly smaller exponents (because $r < d$ and $t > t_1$). Since $E_{e_1}(\mu) = 0$, we get $t_1 + jd \equiv 0 \pmod p$. Put $b_1 = (t_1+dj)/p \in \mathbb{Z}$: $pb_1 > dj$ gives $b_1 \ge m+1$, and $pb_1 \le g' + dj$ gives $b_1 \le R$. Now Φ_{b_1} is purely polar (because $b_1 > m$), with leading term x^{-t_1} (coefficient 1). Replace μ by $\mu - c\,e_{b_1}$ $(c = [x^{-t_1}]\Phi_\mu)$: the part ≥ 0 does not change, μ stays in $\ker E$, and t_1 strictly increases. After at most g' steps, $t_1 > g'$: we have $\mu = \mu^{\mathrm{fin}} + \sum c_b e_b$ with $\mu^{\mathrm{fin}} \in \ker C$ and $b > m$.

$\ker C \cap X_{>m} = 0$. For $m < b \le R$, row b of B is zero for $t < t_b := pb - dj$ (exponent $> dj = \deg f^j$) and equals 1 at $t = t_b$ (f monic); moreover $1 \le t_b \le g'$ (because $b > m$ and $b \le R$) and t_b is strictly increasing: these rows are independent.

Conclusion. $\dim\ker E = \dim\ker C + (R-m)$, hence $\kappa = \dim\ker C = \gamma$. □

Corollary 11. *(i)* $\operatorname{rank} M^{(k)} - \operatorname{rank} M^{(n-k)} = g_k + g_{k_{\mathrm{out}}} - (d-1)$. *(ii) For a fixed point $k_{\mathrm{out}} = k$: $m \le d-1-g_k$. (iii) If $k_1 = n-k_0$ and $k_{0,\mathrm{out}} = k_1$ (self-dual 2-cycle), the two blocks have the same rank.*

Proof. (i) $\kappa = g_k - \operatorname{rank} M^{(k)}$ and $\gamma = g'_{\mathrm{out}} - \operatorname{rank} M^{(n-k)}$ with $g'_{\mathrm{out}} = d-1-g_{k_{\mathrm{out}}}$ (Lemma 9); apply Theorem 10. (ii) $m \le R = d-1-g_{k_{\mathrm{out}}}$ (Lemma 9(d),(e)). (iii) Here $g_{k_0} + g_{k_1} = d-1$. □

4. What is now proved, and what is not

Proved (for f squarefree, $\gcd(d,n) = 1$, $p \nmid n$): Theorem 2 (which fills the gap "poles of order $\ge p$"), Lemma 3, Proposition 6 ($\dim S = m = \lfloor dj/p \rfloor$, which shows that $s_0 \equiv dj$ adds nothing), Theorem 7 (exact formula), Corollary 8, Theorem 10 (K) and Corollary 11.

Not proved: (a) everything concerning cycles of length ≥ 2 (the theorem (K) does not extend to composites; the intersection $\mathrm{im} M_0 \cap \ker M_1 \ne 0$ of the 70 pure cases remains an observation); (b) $\gcd(d,n) > 1$; (c) σ with f not squarefree; (d) the rule $t\,r \equiv \mathrm{gap} \cdot p \pmod d$ for pairs of proportional rows; (e) the 7 dependencies among ≥ 3 rows.

Numerical control checks (independent of the proofs): Theorem 2/Corollary 4 on 6 175 cases (0 discrepancy), $\dim S = m$ on 2 074 cases, Lemma 9 on 7 370 triples (n,p,d,k), and (K) in Sage on 61 658 cases (previous note).